

ANKIT KUMAR

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Education

Indian Institute of Technology, Kanpur

Master of Technology (M.Tech), Department of Management Sciences (DoMS)

Kanpur, India
July 2025 – Present • CPI: **7.18 / 10.0**

C.S.A. University of Agriculture and Technology

Bachelor of Technology (B.Tech) in Agricultural Engineering Aug 2021 – June 2025 • CPI: **8.25 / 10.0** (*Chancellor's Silver Medalist*) Kanpur, India

Professional Experience

Tata Consultancy Services (TCS)

Data Science & AI Intern | Computer Vision & Machine Learning

Remote / India
May 2026 – July 2026

- Engineered an automated vehicle CO₂ emission tracking pipeline from traffic CCTV feeds for enterprise ESG reporting.
- Implemented **YOLOv8** object localization paired with **OpenCV** image preprocessing and **EasyOCR** with regex syntax correction.
- Trained dual-stage ensemble estimators: **Gradient Boosting** ($R^2 = 0.996$, RMSE = **3.12** g/km) and Random Forest; delivered interactive **Streamlit** dashboard with **17-test Pytest** suite.

Key Flagship Projects: Machine Learning, Forecasting & Deep Learning

Petrol Price Forecasting & Analytics System

Academic Project (Guide: Prof. Faiz Hamid, IIT Kanpur) | PyTorch LSTM, XGBoost, ARIMA

Aug 2025 – Nov 2025

[\[GitHub Repo\]](#)

- Built a multi-model time-series forecasting pipeline predicting weekly fuel pricing using **PyTorch LSTM**, **XGBoost**, and **ARIMA(5,1,0)**.
- Engineered **15+ temporal features** (1–12 lags, 4-wk/12-wk SMA/EMA, rolling volatility); tuned XGBoost delivered **1.12% MAPE** on 35% unseen test data.

Credit Risk Modelling: Loan Eligibility Prediction

Academic Project (Guide: Prof. Faiz Hamid, IIT Kanpur) | Scikit-learn, XGBoost, SMOTE

Aug 2025 – Nov 2025

[\[GitHub Repo\]](#)

- Constructed an enterprise credit scoring pipeline analyzing **51,336 customer records** with 79 attributes to predict default risk.
- Conducted statistical feature selection via **Chi-Square**, **ANOVA**, and **VIF**; rebalanced data with **SMOTE**; achieved **78% test accuracy** via **XGBoost**.

Solar Panel Defect Classification

Academic Project (Guide: Prof. Veena Bansal, IIT Kanpur) | TensorFlow, MobileNetV2

Jan 2026 – May 2026

[\[GitHub Repo\]](#)

- Trained a 6-class deep learning defect detection pipeline on **885 electroluminescence images** using **TensorFlow** transfer learning.
- Applied Batch Normalization, Dropout (0.3), and L2 decay; fine-tuned **MobileNetV2**, attaining **82.5% validation accuracy**; deployed via Streamlit.

Additional Relevant Machine Learning & Analytics Projects (Full Portfolio)

- Cancer Risk Level 3-Class Stratification:** Scikit-learn, Random Forest, Gradient Boosting across 23 biomarkers (**>98% accuracy**).
- Breast Cancer Biopsy Malignancy Prediction:** Wisconsin FNA clinical dataset, PCA dimensionality reduction, ROC-AUC model benchmarking.
- Crop Yield End-to-End AgriTech ML Pipeline:** Feature transformers, climate/pesticide regressors, interactive Streamlit simulation app.
- Telecom Churn Prediction & Customer Analytics:** Cost-sensitive subscriber retention modeling, high-recall optimization, Pandas & Seaborn.
- Smart Appliance Energy Consumption Prediction:** Multi-room IoT sensor analytics, ensemble regressors for low-energy building efficiency.
- City Municipal Payroll Analytics:** Parametric/non-parametric hypothesis testing, Two-Way ANOVA across civic departments.
- NVIDIA Stock Quantitative Trajectory Prediction:** Technical indicators (MACD, RSI, Bollinger Bands, ATR) benchmarked against random walk.
- Electricity Demand Time-Series Forecasting:** Stationarity testing (ADF, KPSS), seasonal differencing, SARIMAX vs recurrent LSTM.

Technical Skills, Coursework & Certifications

- ML & Deep Learning:** Scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch, TensorFlow, Keras, OpenCV, YOLOv8, CNN, LSTM, Optuna, SMOTE
- Statistical Methods & Tools:** Python (Advanced), SQL (PostgreSQL, MySQL), Pandas, NumPy, SciPy, Statsmodels, A/B Testing, ANOVA, FastAPI, Docker
- Relevant Courses (IIT Kanpur):** Probability & Statistics, Applied Machine Learning, Statistical Modelling for Business Analytics (SMBA), Data Mining
- Honors & Certifications:** **GATE 2025: 98.48 %ile** • **Chancellor's Silver Medal** • **Univ of Tokyo GCI AI** • **Oracle Certified DS & AI**